

WEEK 2 - SEMINAR

PROBABILITY

If you choose an answer to this question at random, what is the chance you will be correct?

A) 25%

B) 50%

C) 60%

D) 25%

RECALL FROM LECTURES

$$\text{PROBABILITY } (P(\text{WANTED})) = \frac{n_{\text{wanted}}}{N \text{ of all possible}}$$

- **CONDITIONAL PROBABILITY:** $P(A | B)$ IS THE PROBABILITY EVENT **A** TO HAPPEN **GIVEN** THAT **B** HAS ALREADY **HAPPENED**.
- **AND:** $P(A \text{ AND } B) = P(A | B)P(B)$ AND $P(B \text{ AND } A) = P(B | A)P(A)$
 - THEN: **$P(A | B)P(B) = P(B | A)P(A)$** (BAYES' THEOREM)
 - IF A,B: **INDEPENDENT**, THEN $P(A \text{ AND } B) = P(B \text{ AND } A) = P(A)P(B)$
- **OR:** $P(A \text{ OR } B) = P(A) + P(B) - P(A \text{ AND } B)$
 - IF A,B: **MUTUALLY EXCLUSIVE** ($P(A \text{ AND } B)=0$), THEN $P(A \text{ OR } B) = P(A) + P(B)$
- EXPECTATION: **$E(X) = \sum px$**

QUESTION 1

A JAR CONTAINS 49 GREEN MARBLES AND 73 BLUE ONES. WHAT'S $P(\text{GREEN})$?

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ANSWER

REMEMBER, A PROBABILITY IS JUST THE FRACTION OF WANTED RESULTS OUT OF POSSIBLE RESULTS.

$$P(\text{GREEN}) = \frac{n_{\text{green}}}{N} = \frac{49}{122} = 0.40$$

QUESTION 2

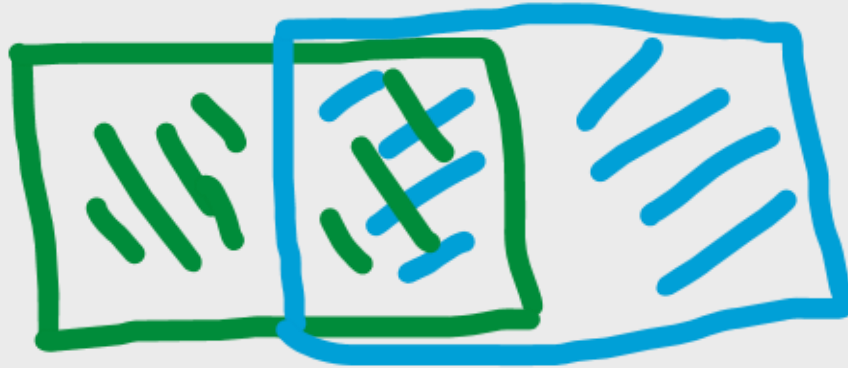
A JAR CONTAINS 77 MARBLES THAT ARE JUST GREEN, 94 THAT ARE JUST BLUE, AND 90 THAT ARE BOTH GREEN AND BLUE. WHAT IS $P(\text{GREEN} \mid \text{BLUE})$?

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ANSWER

REMEMBER, A CONDITIONAL PROBABILITY IS CALCULATED JUST LIKE A SIMPLE ONE - WANTED RESULTS AS A FRACTION OF POSSIBLE RESULTS; WE JUST RESTRICT "POSSIBLE" A BIT WITH THE KNOWLEDGE WE'RE GIVEN.



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$$P(\text{GREEN} \mid \text{BLUE}) = \frac{n_{\text{green in blues}}}{N \text{ of all with some blue}} = \frac{90}{184} = 0.49$$

QUESTION 3

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ANSWER

WE SHOULD USE THE CONDITIONAL PROBABILITY.

$$P(A \text{ AND } B) = P(A | B)P(B) =$$

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ANSWER

WE SHOULD USE THE CONDITIONAL PROBABILITY.

$$P(A \text{ AND } B) = P(A | B)P(B) = 0.53 * 0.32 = 0.17$$

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BONUS QUESTIONS:

1. CALCULATE $P(B | A)$
2. CALCULATE $P(A \text{ AND } B)$ ASSUMING A,B INDEPENDENT
3. CALCULATE $P(A \text{ AND } B)$ IF A,B MUTUALLY EXCLUSIVE

QUESTION 3

1. CALCULATE $P(B | A) = P(A \text{ AND } B) / P(A) = 0.17 / 0.96 = 0.18$
2. CALCULATE $P(A \text{ AND } B)$ ASSUMING A,B INDEPENDENT
$$P(A \text{ AND } B) = P(A) * P(B) = 0.96 * 0.32 = 0.31$$
3. CALCULATE $P(A \text{ AND } B)$ IF A,B MUTUALLY EXCLUSIVE 0

QUESTION 4

A SURVEY PRODUCES THE FOLLOWING RESULTS:

	TEA	COFFEE	HOT CHOCOLATE
UG	69	5	76
PG	93	44	96
STAFF	55	31	64

CALCULATE THE $P(\text{TEA})$.

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CALCULATE THE $P(\text{TEA})$.

ANSWER

THE WAY TO CALCULATE A SIMPLE PROBABILITY FROM A TABLE IS TO ADD UP THE RESULTS YOU WANT, AND DIVIDE BY THE TOTAL NUMBER.

HOW MANY DRINK TEA? $69+93+55 = 217$

THEN, $P(\text{TEA}) = 217 / 533 = 0.41$

QUESTION 5

A SURVEY PRODUCES THE FOLLOWING RESULTS:

	TEA	COFFEE	HOT CHOCOLATE
UG	65	43	36
PG	5	26	18
STAFF	56	79	30

CALCULATE THE $P(\text{UG} \mid \text{TEA})$.

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CALCULATE THE $P(\text{UG} | \text{TEA})$.

ANSWER

THE WAY TO CALCULATE A CONDITIONAL PROBABILITY FROM A TABLE IS TO ADD UP THE RESULTS YOU WANT, AND DIVIDE BY THE TOTAL OF THE GIVEN

HOW MANY DRINK TEA? $65+5+56 = 126$ (GIVEN)

HOW MANY ARE UG FROM THE GIVEN ? 65

THEN, $P(\text{UG} | \text{TEA}) = 65 / 126 = 0.52$

QUESTION 6

THE PREVALENCE OF A DISEASE IS 23%. A TEST FOR THAT DISEASE IS 77% SENSITIVE AND 55% SPECIFIC.

IF SOMEONE IS TOLD THE TEST CAME BACK POSITIVE, WHAT IS THE PROBABILITY THEY DO ACTUALLY HAVE THE DISEASE?

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☐ $P(D) = 0.23$

☐ $P(P | D) = 0.77$

☐ $P(N | ND) = 0.55$

☐ $P(D | P) = ?$

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➤ $P(P) = P(P | D)P(D) + P(P | ND)P(ND) = 0.77*0.23 + (1 - P(N | ND))(1 - P(D))$
 $= 0.177 + 0.346 = 0.523$

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QUESTION 7

THE PAYOFFS FOR A PORTFOLIO IN THREE CIRCUMSTANCES ARE 91, 29, AND 29.

AN ANALYST ESTIMATES THE PROBABILITIES OF THE FIRST TWO CIRCUMSTANCES AS 0.09 AND 0.27 RESPECTIVELY, WITH THE THIRD CIRCUMSTANCE HAPPENING THE REST OF THE TIME.

WHAT IS THE EXPECTED PAYOFF OF THE PORTFOLIO?

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ANSWER

THE EXPECTED AMOUNT IS THE SUM OF EACH OUTCOME WEIGHTED BY (MULTIPLIED BY) THE PROBABILITY OF THAT OUTCOME

$$E(X) = \sum p_x = (0.09*91) + (0.27*29) + ((1 - 0.09 - 0.27)*29) = 34.58$$

QUESTION 8

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$$C(n) = \frac{n!}{r!(n-r)!} = 2,349,060$$

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IF INSTEAD OF A COMMITTEE WHERE EVERYONE IS EQUAL, EACH OF THE 5 PEOPLE CHOSEN FROM 51 HAS A DIFFERENT JOB, HOW MANY WAYS TO CHOOSE THE JOBS ARE THERE?

ANSWER

SINCE EACH HAS A DIFFERENT JOB, ORDER IS IMPORTANT.

$$P(n) = \frac{n!}{(n-r)!} = \mathbf{281,887,200}$$

- IF THE ORDER **WAS NOT** IMPORTANT (SAME JOB):

$$C(n) = \frac{n!}{r!(n-r)!} = \mathbf{2,349,060}$$

SUM-UP

1. SIMPLE PROBABILITY
2. CONDITIONAL PROBABILITY
3. AND/OR
4. BAYES' RULE
5. EXPECTATIONS

ANY QUESTIONS?